
Data Visualization with Tableau
Introduction & History
Definition
Graphical presentation of data
Aids quick understanding
Importance
Identifies trends and patterns
Detects outliers/anomalies
Supports decision-making
Pioneers
William Playfair (Bar/Pie charts)
Florence Nightingale (Coxcomb chart)
John Snow (Cholera map)
Charles Minard (Flow maps)
Common Charts
Bar chart (Comparisons)
Line chart (Trends)
Pie chart (Proportions)
Scatter plot (Relationships)
Data Acquisition
Connectors
File-based (Excel, CSV, JSON)
Database (MySQL, SQL Server)
Cloud (BigQuery, Snowflake)
ODBC/JDBC standards
Data Preparation
Data Interpreter (Cleaning)
Custom SQL
Rserve (R communication)
Database Types
SQL (Structured/Fixed schema)
NoSQL (Unstructured/Flexible schema)
Connection Modes
Live Connection
Real-time updates
Queries original database
No duplicate storage
Extracts
.hyper file format
In-memory processing (RAM)
Offline analysis capability
Full vs Incremental refresh
Data Modeling
Layer Components
Logical vs Physical tables
Fact tables (Measurements)
Dimension tables (Descriptions)
Combination Methods
Relationships (Logical link)
Joins (Physical combination)
Unions (Append rows)
Data Blending (Cross-source)
Field Types
Dimensions (Blue/Qualitative)
Measures (Green/Quantitative)
Discrete vs Continuous
File & Logic Systems
Tableau Workbooks
.twb (XML definition only)
.twbx (Packaged with data)
Data Source Files
.tds (Connection definition)
.tdsx (Packaged with extract)
Story
Narrative sequence of views
Workspace Features
Panes
Data Pane (Fields/Sets/Groups)
Analytics Pane (Stats/Forecasting)
Shelves & Cards
Marks Card (Visual attributes)
Rows/Columns shelves
Pages shelf
Analysis Tools
Show Me (Chart recommendations)
Parameters (User variables)
Calculated fields
Context filters